ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** time period	Time period is measured.
2	**(d)** 20 degrees C per minute	Rise of 20 each minute.
3	**(a)** check the experimental method	Several outliers suggest a faulty procedure.
4	**(a)** length of the string	Only mass is changed.
5	**(a)** cm	Height is length.
6	**(b)** height and time	Height changed, time measured.
7	**(c)** 12.0 s	(12.1 + 11.9 + 12.0)/3 = 12.0.
8	**(d)** thickness of dough	It is kept same.
9	**(c)** type of soil	Only the soil type is changed.
10	**(c)** 1.8 s	36/20 = 1.8 s.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Independent variable is changed by us (1); Dependent variable changes as a result and is measured (1).	1+1
12	Senses can be fooled or are not exact (1); Some things are unsafe to touch or taste (1).	1+1
13	A reading far from others (1); Check for error and repeat the measurement (1).	1+1
14	Sum = 16.6 s (1); Mean = 4.15 s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Change: temperature of water (1); Measure: time taken to dissolve (1); Same: mass of sugar, volume of water, stirring (1).	1+1+1
16	Hand spans differ in size (1); Non-standard units give different values (1); Use a metre scale (standard unit) (1).	1+1+1
17	Why does milk rise on heating? (1); Is it due to bubbles of vapour? (1); Tests can be made with a different liquid (1).	1+1+1
	OR: Mean = 54.6 / 3 = 18.2 s (1); Time period = 18.2 / 10 (1); = 1.82 s (1).	1+1+1
18	A number without unit has no meaning (1); Prevents mix-up between cm and m (1); Allows other people to repeat the work (1).	1+1+1
	OR: Same time makes results comparable (1); Fixed procedure removes bias (1); Others can repeat it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Measure temperature fall with a thermometer (1); Constant: volume, vessel, starting temperature, place (1); Heat each to same temperature (1); Record every minute in a table (1); Draw a graph and conclude (1).	1+1+1+1+1
	OR: Independent: amount of fertiliser (1); Dependent: height of plants (1); Hypothesis: more fertiliser gives more height (1); Keep soil, water, light same (1); Use many plants for each amount (1).	1+1+1+1+1
20	Time period rises with length (1); Rise is not proportional (1); Conclusion: longer pendulum, longer period (1); Constant: mass of bob (1); Constant: angle of swing (1).	1+1+1+1+1
	OR: They show what does not work (1); They help others avoid wasted effort (1); They may lead to new discoveries (1); Selective reporting is dishonest (1); A complete picture is built (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up the **independent** variable (changed by you) with the **dependent** variable (measured).
- Changing more than one factor at a time and still calling the test fair.
- Writing a hypothesis as a question or an opinion instead of a testable statement.
- Taking a single reading and skipping repeats; forgetting to average readings.
- Reading a scale or thermometer from the side (parallax) and omitting units in the answer.

- Drawing a conclusion that the data do not support, or generalising from one trial.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** unreliable	Repeatability gives reliability.
2	**(c)** How long does ice take to melt in a warm room?	Time must be measured.
3	**(a)** line graph	A line graph shows variation with time.
4	**(a)** evidence from tests	It is evidence based.
5	**(a)** aim, apparatus, procedure, observations, conclusion	Standard reporting order.
6	**(c)** 1 mm	Least count = 1 mm.
7	**(a)** 7	35/5 = 7.
8	**(b)** zero error	Should read 0 degrees C.
9	**(b)** not a fair test	Two factors are changed.
10	**(a)** compare methods and repeat	Find the reason for difference.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To compare with the changed set (1); To confirm that the change in variable caused the effect (1).	1+1
12	Only testable ideas can be supported or rejected (1); 'This colour is lucky' cannot be tested (1).	1+1
13	Time period = total time / number of swings (1); 27 / 15 = 1.8 s (1).	1+1
14	Measurement uses an instrument and gives a number (1); Observation may be through senses and may be qualitative (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Observation raises a question (1); Hypothesis is tested by an experiment (1); Conclusion is drawn from data and can raise new questions (1).	1+1+1
16	Place it on a level surface (1); Eye in line with the bottom of the meniscus (1); Note the least count (1).	1+1+1
17	Show apparatus set-up (1); Remove confusion in words (1); Help others to repeat (1).	1+1+1
	OR: Observation: milk is rising in the pan (1); Inference: it is about to boil over (1); Inference may be wrong, so it needs checking (1).	1+1+1
18	Repeated tests on many people (1); A comparison group without the medicine (1); Results reviewed by experts (1).	1+1+1
	OR: Start and stop at the right moments (1); Repeat and average (1); Use the same person to time (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Accuracy: closeness to the true value (1); Precision: closeness of repeated readings (1); Example: readings 150.2, 150.3, 150.2 cm are precise (1); If true value is 152 cm, they are not accurate (1); A good measurement needs both (1).	1+1+1+1+1
	OR: Standard units are agreed all over the world (1); Allows comparison of results (1); Hand-span, foot differ among people (1); Trade and science would be confused (1); e.g. fabric length measured in own spans (1).	1+1+1+1+1
20	Fair test: cause is clear (1); Repetition: reliable (1); Tabulation: organised (1); Graph: shows trend (1); Honest reporting: trust (1).	1+1+1+1+1
	OR: Collect data at 20, 40, 60 and 80 degrees C (1); Record in a table with units (1); Plot temperature on x-axis, time on y-axis (1); Join points with a smooth line (1); Read the trend: time falls as temperature rises (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** an over-generalisation from few examples	Few examples are not enough.
2	**(b)** Salt dissolves faster in hot water than in cold water.	It can be tested by measurement.
3	**(a)** parallax error	Eye not level with the mark.
4	**(b)** a bar graph	Categories compared in bars.
5	**(d)** Time taken (s)	Quantity with unit is stated.
6	**(d)** The soil is wet, so it rained.	It goes beyond what is observed.
7	**(c)** personal opinion of the investigator	Procedure and data are what matter.
8	**(d)** a value far from the others	It differs greatly.
9	**(c)** compare the changed set with an unchanged set	It shows the baseline.
10	**(d)** recording errors	Record immediately.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vapours catch fire easily (1); May cause serious burns or explosion (1).	1+1
12	Instrument does not read zero when it should (1); It must be subtracted or corrected from the readings (1).	1+1
13	Data become organised (1); Plotted points can be taken easily (1).	1+1
14	Show trend of change with time (1); Predict values between and beyond readings (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Chance and special conditions may affect the result (1); e.g. one sunny day cannot show that plants grow faster in sunlight (1); Repeat with many samples (1).	1+1+1
16	Bar graph: compare categories (1); Line graph: change over time (1); Pie chart: parts of a whole (1).	1+1+1
17	Results often leave gaps (1); New questions arise from them (1); These start fresh investigations (1).	1+1+1
	OR: Most points show a smooth trend (1); A stray point lies far away (1); It signals a reading to check or repeat (1).	1+1+1
18	A hypothesis is an explanation (1); A prediction states the expected result if the hypothesis is true (1); Both can be tested (1).	1+1+1
	OR: Other factors, like warmth or water, may differ (1); Control test needed with equal conditions (1); So the claim is not yet proved (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Independent: light (1); Dependent: growth/colour of the plant (1); Constants: water, soil, plant type (1); Expected: best growth in sunlight (1); Dark plant is pale and weak (1).	1+1+1+1+1
	OR: Collect facts before deciding (1); Test ideas on a small scale (1); e.g. check which detergent cleans better by a fair test (1); e.g. test whether a bulb or wire has fault by trial (1); Evidence-based decisions are better (1).	1+1+1+1+1
20	Bulb touching the vessel gives a wrong value (1); Not waiting for a steady reading (1); Parallax in reading (1); Zero error of instrument (1); Handle carefully; read at eye level (1).	1+1+1+1+1
	OR: It narrows down the possible explanations (1); Fresh hypotheses can be formed (1); Example: if seeds do not sprout without light, test water (1); Failure may point to hidden factors (1); Progress comes from corrected ideas (1).	1+1+1+1+1

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